

## CLINICAL GUIDANCE



# Multi-disciplinary collaborative consensus guidance statement on the assessment and treatment of cognitive symptoms in patients with post-acute sequelae of SARS-CoV-2 infection (PASC)

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## INTRODUCTION

The emergence of coronavirus disease 2019 (COVID-19) caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) has brought with it a plethora of new challenges. In the beginning of the pandemic, efforts were focused on pathogenesis and acute treatment; however, over time, understanding and managing post-COVID sequelae have become the new frontier.<sup>1,2</sup> Generally, the majority of individuals show symptom resolution within 3–4 weeks of COVID-19, but a substantial number of people

continue to experience lingering effects and develop protracted illness, regardless of initial symptom severity. Although still being defined, these effects can be collectively referred to as postacute sequelae of SARS-CoV-2 infection (PASC),<sup>3</sup> which is the term used in this report. Notably, there are a number of other terms that are found in the literature (eg, long COVID, postacute COVID-19 syndrome, long-haul COVID, chronic COVID). At the time of development, much of the literature focused on patients who were not vaccinated, and the incidence and trajectory of PASC in vaccinated patients with

“breakthrough” cases (including but not limited to current and emerging variants of the virus) are evolving. The PASC Collaborative took this into account during the development process and these guidance statements generally apply to individuals who develop PASC regardless of their vaccination status.

This guidance statement has a specific focus on the cognitive-related symptoms of PASC that can occur in people who have been diagnosed with acute COVID-19 infection or presumed to have had the infection and initially experienced mild to severe symptoms. Some patients required hospital acute care, whereas many others were managed in nonhospitalized community settings. This consensus guidance statement is one in a series extending across the breadth of the most prevalent or recognized PASC sequelae. Published and in-process guidance statements from this collaborative include the assessment and management of PASC associated fatigue, breathing and respiratory sequelae, cardiovascular complications, autonomic dysfunction, mental health, and neurologic sequelae. These statements are intended to provide consensus-driven practice guidance to clinicians in the assessment and treatment of individuals presenting with PASC.

At the time of this report, approximately 5.2 million people worldwide have died from COVID-19 and 235 million people have “recovered;”<sup>4</sup> however, the absolute percentage of individuals who have completely recovered remains evasive. Fatigue, dyspnea, cough, anosmia, cognitive symptoms (“brain fog”), and dysgeusia are the most common symptoms encountered in PASC.<sup>2</sup> Numerous studies have demonstrated disproportionately high infection or morbidity and mortality rates in certain populations including, but not limited to, racial/ethnic minority groups and people who live or work in prisons and other detention centers. Some studies suggest that women may be at higher risk for developing PASC-related symptoms.<sup>5</sup>

It is widely acknowledged that systematic study of PASC is needed to develop an evidence-based approach for caring for individuals with PASC. At present, there is a dearth of rigorous scientific evidence regarding effective assessment and treatment of PASC that prevents the creation of evidence-based clinical guidelines. However, the U.S. health care system is seeing an increase in the number of patients presenting with PASC and there is an urgent need for clinical guidance in treating these patients.

As these Consensus Guidance Statements on PASC are published they will be available via <https://onlinelibrary.wiley.com/>.

## PASC CONSENSUS GUIDANCE STATEMENT METHODS

The American Academy of Physical Medicine and Rehabilitation (AAPM&R) Multi-Disciplinary PASC

Collaborative (PASC Collaborative) was created, in part, to develop expert recommendations and guidance from established PASC centers with extensive experience in managing patients with PASC. The PASC Collaborative is following an iterative, development approach to achieve consensus on assessment and treatment recommendations for a series of Consensus Guidance Statements focused on the most prominent PASC symptoms. These statements were developed by a diverse team of experts, with input from patient representatives with a history of PASC. These statements integrate evolving experience and expertise with available evidence to provide tools to clinicians treating patients. There is an intentional focus on health equity as disparities in care and outcomes are critically important to address. Beyond patient care, the hope is that a broadened understanding of current patient care practices will help identify areas of future research. A full description of the methodology has also been published.<sup>6</sup>

We acknowledge that the definition of PASC is evolving, and there are various factors that contribute to both diagnosis and treatment. For example, previous literature has suggested that PASC be defined as the continuation of symptoms beyond 3 or 4 weeks from the onset of acute infection.<sup>7</sup> Other definitions of PASC include symptoms lasting longer than 3 months.<sup>8</sup> Based on feedback from clinicians and patient representatives that earlier evaluation, diagnosis, and management can improve access to beneficial interventions, for the purpose of this Consensus Guidance Statement, we recommend expanded PASC assessment if symptoms are not improving 1 month after acute symptom onset.

These Consensus Guidance Statements are intended to reflect current practice in patient assessment, testing, and treatments. They should not preclude clinical judgment and must be applied in the context of the specific patient, with adjustments for patient preferences, comorbidities, and other factors.

## COGNITIVE SYMPTOMS IN INDIVIDUALS WITH PASC

Prominent neurological and behavioral symptoms have been reported in the growing PASC literature. Common neurological and neuropsychiatric symptoms in individuals with PASC include fatigue, myalgia, headaches, sleep disturbance, anxiety, depression, dizziness, anosmia, dysgeusia, and cognitive symptoms, often called a “brain fog.” It is important for clinicians to recognize that disease severity may not be a predictor of PASC symptoms as many patients presenting to outpatient COVID recovery centers experienced only mild initial SARS-CoV-2 infection. Primary cognitive symptoms include deficits in reasoning, problem solving, spatial planning, working memory, difficulty with word retrieval,

and poor attention.<sup>9-12</sup> In addition, small studies in patients recovering from COVID-19 who develop postural orthostatic tachycardia syndrome (POTS) have shown worsening executive function and attention in the standing position.<sup>13</sup> Assessment and treatment of cognitive symptoms in patients with PASC is the focus of this review.

The physiologic effects of SARS-CoV-2 infection on the central nervous system (CNS) are incompletely understood. Neurons express surface angiotensin-converting enzyme-2 (ACE-2) receptors, although at a lesser density than respiratory or gastrointestinal epithelium. Prior human coronaviruses (SARS-CoV-1, Middle East respiratory syndrome) infections have demonstrated neurotropism, invading the CNS via ACE-2 receptor binding and migration along motor and sensory pathways. Experimental intranasal SARS-CoV-1 in mice has demonstrated transmission along the olfactory bulb, with subsequent transmission to limbic structures, and progression to brainstem and spinal cord.<sup>14</sup> Reporting to date is not convincing that SARS-CoV-2 infection in humans typically involves CNS infection, even in cases of severe respiratory infection. Studies have included cerebral spinal fluid sampling, brain imaging, and brain examination at autopsy and have demonstrated SARS-CoV-2 within vessels and cerebral support structures but no evidence of intrathecal viral replication.<sup>15</sup>

SARS-CoV-2 infection related ischemic and hemorrhagic stroke, postinfectious encephalitis, acute necrotizing encephalopathy, and endotheliitis have been described but are infrequent.<sup>16</sup> Microthrombi in cerebral microvasculature, central venous thrombosis, and patterns of hypoxic ischemic injury have also been demonstrated. Generalized, focal, and subclinical seizures have been observed.<sup>16</sup> Neurological conditions associated with SARS-CoV-2 infection will be addressed in a future AAPM&R PASC Consensus Guidance Statement.

Neurological symptoms are present in more than 80% of hospitalized patients during the acute phase of the infection. The most frequent neurologic symptoms persisting after the acute onset of the infection were: "brain fog" (81%), headache (68%), numbness/tingling (60%), dysgeusia (59%), anosmia (55%), and myalgias (55%).<sup>17</sup>

Acute toxic metabolic encephalopathy is the most common neurological complication among acutely hospitalized patients with COVID-19 and its etiology is multifactorial including potential contributions from sepsis, hypoxia, hypercarbia, hyponatremia, hyponatremia, uremia, multiorgan system failure, and hyperinflammatory state. Neurologic impairment is more prominent in severe COVID-19 infection. Hospitalized patients with encephalopathy demonstrated significantly increased 30-day mortality.<sup>16,18</sup>

Proposed mechanisms of SARS-CoV-2 related CNS pathophysiology include (1) inflammation-mediated

increased permeability of blood brain barrier (including thinning of the basal lamina) with activation of CNS immune mediators and glia; (2) an associated activation of metabolic pathways to induce synaptic pruning and neuronal loss; (3) excessive glutamate/N-methyl-D-aspartate excitotoxicity and neurotransmitter depletion; (4) subsequent diaschisis effects; (5) unmasking of previous subclinical neurological and neuropsychiatric impairments; and less likely (6) direct SARS-CoV-2 infection.<sup>19</sup>

The aggregate clinical presentation of PASC has features similar to other postinfectious syndromes, including Epstein-Barr, Lyme, Zika, and clinical conditions such as myalgic encephalomyelitis/chronic fatigue syndrome, fibromyalgia syndrome, postintensive care syndrome, and POTS. The many similarities of these conditions including cognitive symptoms, fatigue, headaches, sleep disturbance, and mood disorders may represent a final common pathway for a phenotype of postinfectious immune and neuronal dysregulation. The established literature for these conditions, as well as references for the management of mild acquired brain injury, may be helpful in implementing approaches to treatment and patient education.

An important group of SARS-CoV-2 infected patients to consider are older individuals as there are a number of cognitive-related concerns in this population. (See Appendix: Health Equity Considerations and Examples in Post-Acute Sequelae of SARS-CoV-2 Infection [PASC]: Cognitive Symptoms.) One report noted that healthy older individuals as well as those with mild cognitive symptoms and dementia have been among the most affected with acute COVID-19 infections, and both direct and indirect cognitive-related issues have been described.<sup>20</sup> The report also highlighted that both the patient and the patient's caregiver may experience SARS-CoV-2 related cognitive decline and that this population may have more challenges with virtual care. Worsening conditions, particularly cancer, time-dependent diseases, and degenerative conditions were also noted to be a concern as older individuals who may not have received the routine care they needed during the pandemic. In all age groups, mental health conditions and substance use (e.g., alcohol, recreational drug use) have increased during the pandemic and may additionally contribute to cognitive dysfunction.<sup>21</sup> Preexisting static conditions such as cerebral palsy, stroke, and traumatic brain injury as well as progressive neurologic illnesses such as multiple sclerosis and dementia are important to consider. In female adults, pregnancy and the postpartum period are well documented to cause fatigue and cognitive symptoms and in a subset of patients postpartum depression, which can make the PASC-related diagnosis more challenging. Overlapping fatigue, nonrestorative sleep, dysautonomia, the psychosocial impact of surviving life-threatening illness, and personal losses due to COVID-19, compounded by multifaceted impacts of pandemic-related social isolation,

may further exacerbate cognitive deficits. It is also important to consider preexisting conditions and symptoms to avoid clinical attribution errors.

## PASC-RELATED COGNITIVE SYMPTOM ASSESSMENT RECOMMENDATIONS

As noted in the AAPM&R Multi-Disciplinary PASC Consensus Guidance Statement methodology,<sup>6</sup> the recommendations that follow (Table 1: Cognitive Symptom Assessment Recommendations for Individuals with PASC) are based on expert consensus. Specific guidance recommendations that have been approved by consensus will be noted in the tables and recommendations will be followed by additional discussion.

### Cognitive symptom assessment recommendation statements discussion

Standardized general cognitive screening assessments form the foundation for initial neurocognitive assessment. Commonly used instruments include the Mini Mental Status Examination, Montreal Cognitive Assessment, Saint Louis Missouri Mental Status Exam, Mini-Cog, and the Short Test of Mental Status. These screening instruments are defined in Table 2 and can be administered by a variety of clinical professionals, including primary care. Subsequent formal neurocognitive tools, outlined in Table 4, are specific to those trained in neurocognitive testing and interpretation. As noted, there are no consistent evidence or guideline statements on when to diagnose cognitive symptoms during the SARS-CoV-2 acute course, or after primary symptom resolution in PASC, as SARS-CoV-2 related cognitive symptoms may have delayed onset, and may fluctuate or be continually present from the acute infection phase.

Because of the varying presentations of PASC and often a constellation of overlapping symptoms, consideration of additional studies and referral options for common manifestations is warranted. It should be noted that cognitive screening measures are not a substitute for more in-depth cognitive testing, and abnormal scoring on screening measures does not warrant a cognitive disorder on their own. Table 3 describes diagnostic categories and common findings related to PASC cognitive symptoms, including additional testing and referral options.

### Neurologic deficit assessment

Individuals experiencing PASC may present with a broad constellation of cognitive and neurobehavioral symptoms. It is useful for trained clinicians to sort subjective impairments by cognitive domain and consider

**TABLE 1** PASC cognitive symptom assessment recommendations

| #  | Statement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Patients should be screened for signs of cognitive symptoms using validated tools and instruments. (See Table 2 for commonly used tools/instruments).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2  | Patients should be evaluated for conditions that may exacerbate cognitive symptoms and warrant further testing and potential subspecialty referral (see Table 3). Particular areas include: <ul style="list-style-type: none"> <li>• Sleep impairment</li> <li>• Mood, including anxiety, depression, and posttraumatic stress disorder</li> <li>• Fatigue</li> <li>• Endocrine abnormalities</li> <li>• Autoimmune disorders</li> </ul> Note: Patients often report dissatisfaction with their care because of their persistent symptoms being attributed to psychological factors. It is important to note that mood disorders may be secondary to persistent medical conditions or one of many factors leading to cognitive symptoms.             |
| 3  | Patients should have a thorough neurological examination to identify focal neurological deficits.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 3a | For those patients identified with new or worsening focal neurological deficits (including new or worsening cognitive symptoms) an emergent evaluation is warranted; neuroimaging should be considered.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 4  | The following basic lab workup should be considered to screen for reversible factors contributing to cognitive symptoms. The initial lab workup in new patients or those without lab workup in the 3 months prior to visit including complete blood count, vitamin B12, thiamine, folate, homocysteine, 1,25-dihydroxy vitamin D, magnesium, liver function tests, comprehensive metabolic panel thyroid function tests (thyroid stimulating hormone, free T3, free T4). In high-risk patients, one may consider syphilis rapid plasma regain and human immunodeficiency virus testing. Other laboratory tests may be considered based on the results of these tests or if there is specific concern for comorbid conditions as outlined in Table 3. |
| 5  | Clinicians should conduct a full patient history with review of preexisting conditions and comprehensive medication and supplement review for those that may contribute to cognitive symptoms. Of note, patients with PASC often present on antihistamine, anticholinergic, and antidepressant/anxiolytic medications that can contribute to cognitive symptoms.                                                                                                                                                                                                                                                                                                                                                                                     |
| 5a | Clinicians should validate patient history through the collection of collateral history, including preexisting function and conditions, from care team/primary care, patient family or care partner, or close contact as available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 6  | Clinicians should assess impact of cognitive symptoms using standardized patient-reported assessments, to include activities of daily living, instrumental activities of daily living, school, work and avocational (ie, hobbies), and quality of life.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

Abbreviation: PASC, postacute sequelae of SARS-CoV-2 infection.

**TABLE 2** Cognitive symptom screening tools

| Assessment tool                                                       | Areas of assessment                                                                                            | Administration time          | Pros                                                                                                                                                                                                                                                                                                                                                                                     | Limitations                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Montreal Cognitive Assessment (MoCA) <sup>a</sup>                     | Visuospatial/Executive, Naming, Memory, Attention, Verbal Fluency, Abstraction, Delayed Recall and Orientation | 10 minutes in healthy adults | <ul style="list-style-type: none"> <li>Available in 55 languages</li> <li>Versions for illiterate/low education and blind individuals (can be used in virtual setting)</li> <li>Designed for brief assessment of cognitive impairment across medical/neurological populations</li> <li>Better than MMSE in detecting mild cognitive impairment (MCI)<sup>b</sup> and dementia</li> </ul> | <ul style="list-style-type: none"> <li>Requires training and certification to use</li> <li>Fees required for training, certification and access to assessment tool</li> </ul>                                                                                                                                                                                                                                                                                                |
| Mini-Mental State Examination (MMSE) <sup>b</sup>                     | Orientation, Recognition, Attention, Calculation, Recall and Language                                          | 5 to 10 minutes              | <ul style="list-style-type: none"> <li>One of the most highly used measures in the United States</li> <li>Requires no additional equipment</li> <li>Can provide a method of monitoring deterioration over time</li> </ul>                                                                                                                                                                | <ul style="list-style-type: none"> <li>Biased against people with poor education because of elements of language and mathematical testing</li> <li>Bias against visually impaired</li> <li>Limited examination of visuospatial cognitive ability</li> <li>Poor sensitivity at detecting mild/early dementia</li> <li>Not available via public domain (cost to use)</li> <li>Low ceiling</li> <li>Lack of sensitivity to MCI in young high functioning individuals</li> </ul> |
| Saint Louis University Mental Status Examination (SLUMS) <sup>c</sup> | Orientation, Attention, Calculation, Verbal Fluency, Memory, Clock Drawing and Recognition                     | 7 to 10 minutes              | <ul style="list-style-type: none"> <li>Education-based norms</li> <li>Superior to the MMSE</li> <li>Can detect MCI</li> </ul>                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>No executive components</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                    |
| Mini-Cog <sup>d</sup>                                                 | Memory, Planning                                                                                               | 4 minutes                    | <ul style="list-style-type: none"> <li>Developed and validated for primary care setting</li> <li>Informant/caregiver component</li> <li>Quick measure that does not interfere with work flow</li> </ul>                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>Assesses only very short-term memory and planning</li> </ul>                                                                                                                                                                                                                                                                                                                                                                          |
| Short Test of Mental Status (STMS) <sup>e</sup>                       | Orientation, Recall, Calculation, Abstraction, Information, Construction and Copying, Digit Span               | 5 minutes                    | <ul style="list-style-type: none"> <li>Intended to be more sensitive to problems of learning and mental agility that may be seen in MCI. Incorporates brevity and reasonable sensitivity and specificity for primary care in clinical practice.</li> </ul>                                                                                                                               | <ul style="list-style-type: none"> <li>Fewer items for assessing aspects of executive function as compared to the MoCA</li> </ul>                                                                                                                                                                                                                                                                                                                                            |

<sup>a</sup>MoCA: <https://www.mocatest.org/>.<sup>b</sup>MMSE: <https://www.parinc.com/Products/~//Products/Pkey/238>.<sup>c</sup>SLUMS: [https://www.slu.edu/medicine/internal-medicine/geriatric-medicine/aging-successfully/pdfs/slums\\_form.pdf](https://www.slu.edu/medicine/internal-medicine/geriatric-medicine/aging-successfully/pdfs/slums_form.pdf).<sup>d</sup>Mini-Cog: <https://mini-cog.com/>.<sup>e</sup>STMS: <https://d3hj51r9j54j7.cloudfront.net/wp-content/uploads/sites/5/2011/08/Short-Test-of-Mental-Status.pdf>.



**TABLE 3** Common manifestations, symptoms, additional testing/studies, and referral options for cognitive symptoms

|                                                                        | Neurologic                                                                                                                                                                                                                       | Endocrine                                                                                                                                                                                                                                                                                                                                                                                                        | Autoimmune or infectious                                                                                                                                                                                                                                                                                                                                                                                                               | Mood disorders                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Sleep disorders                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Common symptoms and signs                                              | Symptoms: headache, weakness, numbness, cognitive or communication dysfunction, difficult walking, bowel or bladder dysfunction<br>Signs: focal weakness, abnormal cognitive screening, ataxia, hyperreflexia, aphasia, aprosody | Symptoms: palpitations, fatigue, dizziness, weight gain/loss, sense of chills/fever, irregular menstrual cycle, poor diabetic control, excessive thirst/urination<br>Signs: tachycardia, poor activity tolerance, weight gain/loss, low/elevated temperature, elevated finger-stick/urine glucose, ketotic (fruity) breath, bone pain, risk factors for malnutrition (eg, "tea and toast" diet or alcohol abuse) | Symptoms: rash, joint/muscle pain and stiffness, fever, mouth sores/ulcers, cold/pale/blue/red fingers, sharp chest pain, numbness/tingling/burning in fingers/toes, blurry/decreased vision<br>Signs: rash, arthropathy/swelling/warmth, decreased range of motion, myopathy/tenderness/weakness, fever, Raynaud's phenomena, pleuritic pain on deep breathing, altered sensation, decreased visual acuity                            | Symptoms: anxiety, irritability, chest tightness, low frustration tolerance, depression, fatigue, mood swings, palpitations, change in memory/recall<br>Signs: flat affect/low mood, emotional lability that is, crying/laughing inappropriately, limited impulse control, psychosis                                                                                                                                                                                                                                                         | Symptoms: Poor sleep - hard to get to sleep/wakes frequently/wakes early, nonrestorative/refreshing sleep - "tired" on waking, snoring, frequent urination at night, bad dreams/falls asleep during the day<br>Signs: snoring, restless legs, observed apneic episodes, hypertension, arrhythmias, narcolepsy, congestive heart failure, impaired cognition, poorly controlled mood disorder, metabolic dysfunction: glucose intolerance |
| Further studies to consider in addition to basic laboratory evaluation | For selected cases, neuroimaging including head computerized tomography (CT); MRI, electroencephalography (EEG)                                                                                                                  | Thyroid stimulating hormone (TSH)/free T4 (thyroxine), cortisol levels, growth hormone, luteinizing hormone (LH), follicle stimulating hormone (FSH), testosterone (men), estradiol (women)<br>As in acquired brain injury, new neuroendocrine dysfunction may be present, assess serum cortisol, growth hormone (IGF-I), testosterone, estradiol, prolactin.                                                    | Cytokine storming may induce or unmask autoimmune disease so consider ANA, RF, cyclic citrullinated peptide antibody, HLA B-27 antigen, antithyroid peroxidase antibodies, antiphospholipid antibodies<br>Depending on the patient history, one may add HIV and syphilis (RPR) screening. HSV, EBV, Lyme disease, and Whipple's disease are other less common causes. <sup>54</sup> Consider mast cell activation syndrome assessment. | Screening assessments for behavioral health conditions assist to disentangle psychiatric conditions from cognition impairments and may include assessments of anxiety, depression and posttraumatic stress disorder (PTSD): Hospital Anxiety and Depression Scale (HADS), Beck Depression Inventory (BDI) fast screen; Geriatric Depression Scale (GDS); Patient Health Questionnaire for depression (PHQ-2/9), (or PHQ-8 removing the suicidal thoughts question when immediate review and attention is not available); Generalized Anxiety | Restorative sleep can be assessed using the Epworth Sleepiness Scale (ESS), Stanford Sleepiness Scale, PROMIS Sleep, or Sleep Scale Survey, Insomnia Severity Index screen<br>Sleep apnea evaluation: STOP-BANG questionnaire; overnight sleep study with oximetry to assess for obstructive or central sleep apnea, benefit of CPAP/BIPAP, and oral appliances                                                                          |

(Continues)

TABLE 3 (Continued)

|          | Neurologic                                     | Endocrine     | Autoimmune or infectious            | Mood disorders                             | Sleep disorders                 |
|----------|------------------------------------------------|---------------|-------------------------------------|--------------------------------------------|---------------------------------|
|          |                                                |               |                                     | Disorder 7 (GAD-7);<br>PTSD Checklist-5    |                                 |
| Referral | Neurology, brain injury<br>medicine psychiatry | Endocrinology | Rheumatology,<br>infectious disease | Psychiatry, psychology,<br>neuropsychology | Pulmonology - sleep<br>medicine |

Abbreviations: ANA, antinuclear antibody; CPAP/BIPAP, continuous positive airway pressure/bilevel positive airway pressure; CT, computed tomography; EBV, Epstein-Barr virus; EEG, electroencephalogram; HLA, human leukocyte antigen; HSV, herpes simplex virus; IGF-I, insulin-like growth factor I; MRI, magnetic resonance imaging; PROMIS, Patient-Reported Outcomes Measurement Information System; RF, rheumatoid factor; RPR, rapid plasma reagin.

appropriate formal cognitive assessment tools (see Table 4) and intervention options to form a reasoned treatment plan. It is also important to consider effort testing which can be formally assessed with performance validity measures such as the Test of Memory Malinger, Rey-15, and Advanced Clinical Solutions Word Choice. Such tests aim to detect signs (patterns of test performance) and symptoms (verbal reports - what the patient says), indicative of deliberate distortion and in this context are described as tests of symptom validity. Neurocognitive assessment including performance validity tests should be administered only by trained clinicians such as neuropsychologists who specialize in the administration of neurocognitive tests.<sup>22</sup> Cognitive neuroscience has primarily been conducted in homogenous groups of individuals, and there are calls for greater diversity in study populations to ensure reliability of testing and treatment interventions.<sup>23</sup> Both testing and treatment may be biased or inaccurate for people who identify with racial and ethnic groups or other socially or historically marginalized groups because of a variety of issues such as not having English as a primary language, hearing impairments, not properly accounting for variations in dialect or education level, and stereotype threat. (See Appendix: Health Equity Considerations and Examples in Post-Acute Sequelae of SARS-CoV-2 Infection [PASC]: Cognitive Symptoms.)<sup>24</sup>

If the PASC patient reports primary concerns affecting subjective cognition despite normal performance on screening instruments, referral to neurocognitive rehabilitation specialists or neuropsychologists may be appropriate for more formal and specific assessments. When a clinician without expertise in cognition diagnoses an individual with cognitive impairment, that individual should be referred to a specialist with expertise in cognition for further evaluation and treatment. These practitioners may vary by community but are typically neuropsychologists, speech and language pathologists, or occupational therapists.

It is important to distinguish general or exertional fatigue (Fatigue Severity Scale, Modified Fatigue Impact Scale, Patient-Reported Outcomes Measurement Information System Fatigue) from mental or cognitive fatigue. Mental fatigue is defined as a progressive decrease in cognitive resources over time

when participating in cognitive tasks requiring sustained attention and executive function, independent of deficits from diminished sleep hygiene (daytime sleepiness) or motivation.<sup>25</sup>

Routine neurological physical examination is typically nonfocal for most patients with PASC. The neurologic exam should include assessment of mental status, cranial nerves, motor function, deep tendon reflexes, sensory and coordination function, and evaluation of balance and gait.

When a clinician without expertise in neurological diagnoses is evaluating an individual with neurological impairment, that patient should be referred to brain injury medicine psychiatry or neurology for further evaluation.

In patients with PASC, head computed tomography is typically unrevealing, although contrast studies rarely demonstrate leptomeningeal enhancement.<sup>26</sup> Research-based positron emission tomography scan assessments have revealed hypometabolism in frontal, temporal, and cerebellar regions.<sup>27</sup> Most patients with PASC have had normal brain magnetic resonance imaging (MRI). A study of 48 patients presenting to a neuro-COVID clinic found the most common brain MRI abnormality was nonspecific white matter changes. None of these patients were found to have abnormalities of magnetic resonance (MR) vessel wall imaging.<sup>17</sup>

Persistent headache (duration >4 hours, daily, upon waking)<sup>28</sup> or worsening neurological or cognitive deficits, including headache progression, warrant brain imaging. The SNOOP10 mnemonic can be used as a screening tool in patients with headaches to identify the presence of red flags that would warrant investigation for a secondary cause.<sup>29</sup> Although most headaches are evaluated and managed by primary care providers, consider referral to a specialist if there is uncertainty regarding the diagnosis or management of a patient with headaches.

Ongoing metabolic abnormalities can contribute to cognitive symptoms. Preliminary laboratory tests include: comprehensive metabolic panel including blood urea nitrogen, serum creatinine and liver function, and folate and vitamin B12 levels. Inflammatory mediators may smolder or reactivate in setting of postacute opportunistic infection, warranting reassessment including complete blood count with differential,

**TABLE 4** Neurocognitive assessment tools and therapeutic intervention strategies by cognitive domain

| Cognitive domain*                        | Patient concerns                                                                                                                                                                                                                                                  | Assessment tools                                                                                                                                                                                                                                                                                                                                                                        | Therapeutic interventions                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Attention                                | <ul style="list-style-type: none"> <li>Brain fog</li> <li>Difficulty with concentration on tasks</li> <li>Losing train of thought</li> <li>Misplacing objects</li> <li>Miscalculation</li> <li>Easily distracted</li> </ul>                                       | <ul style="list-style-type: none"> <li>Digit Span</li> <li>Digit Vigilance Test</li> <li>Cancellation</li> </ul>                                                                                                                                                                                                                                                                        | Attention process training for verbal and nonverbal tasks, metacognitive strategies, timed structured activities, minimize distractions.                                                                                                                                                                                                                                                                                                             |
| Processing speed                         | <ul style="list-style-type: none"> <li>Slowed thought processes</li> <li>Difficulty following conversations</li> </ul>                                                                                                                                            | <ul style="list-style-type: none"> <li>Coding</li> <li>Continuous Performance Test</li> <li>Symbol Search</li> <li>Trails A</li> </ul>                                                                                                                                                                                                                                                  | Recording talks, lectures, etc. to review at own pace; practicing skills repeatedly towards automaticity; breaking projects to components to complete over time.                                                                                                                                                                                                                                                                                     |
| Motor function/speed                     | <ul style="list-style-type: none"> <li>Slowed motor function</li> </ul>                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>Grooved Pegboard</li> </ul>                                                                                                                                                                                                                                                                                                                      | Use of dictation devices (speech to text).                                                                                                                                                                                                                                                                                                                                                                                                           |
| Language                                 | <ul style="list-style-type: none"> <li>Word finding difficulty during conversation</li> <li>Grasping for words</li> <li>Verbal fluency</li> <li>Difficulty with comprehension of multiple step instructions due to changes in recall and comprehension</li> </ul> | <ul style="list-style-type: none"> <li>Letter Fluency</li> <li>Category Fluency</li> <li>Boston Naming Test</li> <li>Neuropsychological Assessment Battery (NAB) Naming</li> <li>Multilingual Aphasia Examination (MAE) Token Test</li> <li>Portions of the Western Aphasia Battery-R</li> <li>Scales of Cognitive and Communicative Ability for Neurorehabilitation (SCCAN)</li> </ul> | <p>Semantic feature analysis, word finding strategies, use word associations, convergent/divergent naming tasks, anagrams.</p> <p>Structured tasks with speech-language pathology (SLP) to address various domains, such as comprehension, recall, word finding, thought organization, identification of strategies for all domains that are impaired based on assessments.</p>                                                                      |
| Memory                                   | <ul style="list-style-type: none"> <li>Poor learning and recall of novel information</li> <li>Forgetting daily activities and to do tasks</li> </ul>                                                                                                              | <ul style="list-style-type: none"> <li>California Verbal Learning Test-III</li> <li>Hopkins Verbal Learning Test</li> <li>Rey Auditory Verbal Learning Test</li> <li>Brief Visual Memory Test</li> <li>Rey-Osterrieth Complex Figure Test</li> <li>Wechsler Memory Scale-IV</li> </ul>                                                                                                  | Use of memory strategies during structured tasks, identify strengths (visual vs auditory, note taking, repetition, rehearsal, etc.), use of phone or smartphone applications, calendar, day planner, to do lists.                                                                                                                                                                                                                                    |
| Mental fatigue                           | <ul style="list-style-type: none"> <li>Mental exhaustion/brain fog due to sensory stimulation or after completing cognitive tests for extended periods without breaks</li> </ul>                                                                                  | <ul style="list-style-type: none"> <li>Mental Fatigue Scale<sup>55</sup></li> <li>State Trait Inventory of Cognitive Fatigue (STI-CF)</li> </ul>                                                                                                                                                                                                                                        | <p>Energy Envelope Theory: maintaining expended energy levels consistent with available energy levels may reduce the frequency and severity of symptoms.<sup>56</sup></p> <p>Mindfulness-Based Stress Reduction (MBSR): method of using meditation and yoga to cultivate awareness and reduce stress.<sup>57</sup></p>                                                                                                                               |
| Executive function                       | <ul style="list-style-type: none"> <li>Error monitoring, for example, increased frequency of errors made in work/school performance</li> <li>Trouble planning, organizing and sequencing (e.g., cooking, finances)</li> <li>Difficulty multitasking</li> </ul>    | <ul style="list-style-type: none"> <li>Trails B</li> <li>Stroop Color Word</li> <li>Delis-Kaplan Executive Function Scales</li> <li>Tower of London</li> <li>Wisconsin Card Sorting Test</li> <li>Functional Assessment of Verbal Reasoning and Executive Strategies (FAVRES)</li> </ul>                                                                                                | <p>Training in metacognitive strategies to promote self-awareness and self-monitoring.</p> <p>Examples of metacognitive strategies: Goal-Plan-Do-Review, self-talk, Goal Management Training (Stop-Think - Plan), Predict-Perform Technique.</p> <p>Use of organizational aids (calendar, filing system, daily checklist for chores/tasks, etc.), alarms/reminders on a phone or smartwatch, creating routines, placing items at the same place.</p> |
| Visuospatial or visuoconstruction skills | Blurred vision, depth perception, neglect                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>Line Orientation</li> <li>Figure Copy</li> <li>Clock Drawing Test</li> <li>Rey-Osterrieth Complex Figure Test</li> <li>The Bells Test</li> </ul>                                                                                                                                                                                                 | <p>Use of language-mediated strategies such as self-talk or verbalization to solve problems or remember information.</p> <p>Application of compensatory strategies/aids for writing and organization,</p>                                                                                                                                                                                                                                            |

(Continues)



**TABLE 4** (Continued)

| Cognitive domain* | Patient concerns | Assessment tools | Therapeutic interventions                                                                                                                                                                                                                                                                                                                                 |
|-------------------|------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   |                  |                  | including use of graph or lined paper, limiting problems or text on a page, and highlighting key words or text to remember.<br>Use of visual cancellation tasks, strategies for visual organization such as scanning from left to right, top to bottom for symbols, shapes, numbers, etc.<br>Verbalize maps (eg, use of global positioning system [GPS]). |

\*Influence of effort on subjective cognitive concerns and performance-based neuropsychological test results should be considered, consistent with literature on standardized cognitive assessments following mild traumatic brain injury.<sup>58</sup> All above measures in addition to performance-based validity measured should be administered and interpreted by doctoral-level trained professional with dedicated training in neurocognitive testing.

erythrocyte sedimentation rate, C-reactive protein, ferritin, procalcitonin, urinalysis, and chest x-ray. Inflammatory coagulopathy can be assessed with D-dimer, fibrinogen, and international normalized ratio. For patients with current cardiac symptoms or patients with cardiac issues during acute infection (e.g., myocarditis, heart failure or demand ischemia), additional assessment may include troponins and brain natriuretic peptide. Thyroid function studies, antinuclear antibody, and creatinine kinase may be appropriate in patients with weakness and myalgias.

Although these recommendations draw in part on the American Academy of Neurology's recommendations for reversible dementia,<sup>30</sup> this does not imply that COVID-19 leads to dementia nor that PASC is a progressive neurological condition.

## PASC COGNITIVE SYMPTOM TREATMENT RECOMMENDATIONS

It is important for clinicians to recognize individual patient symptom timelines vary widely and it is possible that initiating treatment earlier will result in earlier resolution of symptoms. The absolute duration of PASC-related cognitive symptoms should not dictate management approach. Alternatively, individual symptom and clinical presentation, preexisting and COVID-19 exacerbated comorbidities, and impacts on function and quality of life should guide an identification and intervention approach (Table 5).

As with any treatment plan, clinicians treating patients with PASC are encouraged to discuss the unknowns of PASC treatments and prognosis, as well as the pros and cons of any therapeutic approach. Patients have reported symptom improvement after following a structured and titrated return to activity.<sup>17</sup> The long-term goal is to progress to resumption of exercise, return to work, and avocational interests. Energy conservation is a key management approach for many patients. A rehabilitation team approach for patients with cognitive issues and

fatigue can be considered with pacing and energy conservation strategies, such as dividing up a longer task into small increments with judicious breaks. For memory and organization issues, application of established techniques used for patients with concussion or traumatic brain injury includes taking notes, using a planner or phone app to record information, and setting electronic reminders. Other tactics include reducing screen time, proper sleep maintenance, managing stress, and increasing physical activity.<sup>31-33</sup>

Based on brain injury research, cognitive rehabilitation is an effective treatment for cognitive symptoms and should be tailored to the individual patient.<sup>34</sup> Interventions should involve a combination of remediation through direct training, metacognitive strategy instruction, and use of compensatory techniques (eg, memory notebooks, alarms, smartphone apps, calendars). Table 4 offers intervention options sorted by cognitive domain.

## Interventions for sleep to improve cognitive symptoms

The detrimental impact of poor sleep on cognition is well established. The ideal duration of total sleep is 7 to 8 hours per night.<sup>35</sup> Treatment approaches include behavioral (nonpharmacologic) and pharmacological and referral to sleep specialists should be considered. Behavioral sleep hygiene strategies include avoiding caffeine, avoiding screen time, avoiding liquids before bed to prevent interruption of sleep with the need to urinate, engaging in relaxation activities, and maintaining a consistent sleep/wake schedule.<sup>36</sup> Treating potential sleep apnea and referral for formal monitored sleep study are recommended as effective sleep apnea treatment can enhance cognition,<sup>37</sup> including attention, processing speed, memory, and executive functioning.<sup>38</sup> Cognitive behavioral therapy for insomnia is recommended as a therapeutic, evidence-based form of treatment for chronic insomnia in adults.<sup>39</sup>

**TABLE 5** PASC cognitive symptom treatment recommendations

| #  | Statement                                                                                                                                                                                                                                                                                                                                                            |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | For patients who screen positive for cognitive symptoms, refer to a specialist (ie, speech-language pathologist, occupational therapist, neuropsychologist) with expertise in formal cognitive assessment and remediation. (See Table 3 for specialist referral options; and Table 4 for examples of Assessment Tools and Intervention Options by Cognitive Domain.) |
| 2  | Treat, in collaboration with appropriate specialists, underlying medical conditions, such as pain, insomnia/sleep disorders (including poor sleep hygiene), and mood disorders that may be contributing to cognitive symptoms.                                                                                                                                       |
| 3  | Complete, in collaboration with patient primary care provider, medication polypharmacy reduction, weaning or deprescribing medications if medically feasible with emphasis on medications that may impact cognition                                                                                                                                                  |
| 4  | Reinforce sleep hygiene techniques including nonpharmacologic approaches as first line of sleep remediation.                                                                                                                                                                                                                                                         |
| 5  | Similar to patients experiencing “physical” fatigue, patients should be advised to begin an individualized and structured, titrated return to activity program.                                                                                                                                                                                                      |
| 5a | For patients who achieve a return to their normal, daily activities, regular exercise (at least 2–3 times/week of aerobic exercise) may be effective in improving cognition and also contribute to improved sleep patterns.                                                                                                                                          |
| 5b | Frequent assessment of the impact of return to normal, daily activities (including school, work, driving, operating heavy machinery, etc.) is recommended to ensure that symptoms do not flare and exercise is tolerated.                                                                                                                                            |

Abbreviation: PASC, postacute sequelae of SARS-CoV-2 infection.

Nutritional supplements may have benefit as subjective sleep quality can be improved by the use of amino acids, melatonin, and Vitamin D.<sup>40–42</sup> There is no indication that magnesium, zinc, resveratrol, or nitrate supplementation are beneficial for restoration of normal sleep.<sup>43</sup>

Should conservative interventions produce incomplete effect, then pharmacological interventions may be indicated. Medications include benzodiazepine receptor agonists, histamine receptor antagonists, melatonin receptor agonists, and dual orexin receptor antagonists. Medications such as benzodiazepines or antihistamines with anticholinergic effects should be prescribed judiciously and usually for short periods of time as they may affect cognition, particularly in older patients. The benefit of prescribing medications to improve sleep should be considered together with this potential risk of cognitive side effects. Sedating medication may be considered by providers if the patient does not have a therapeutic response to first-line medications. These may include medications such as trazodone, mirtazapine, gabapentin, or over-the-counter medications whose active ingredients producing sedating effects (eg, diphenhydramine). It is important to note that none of these medications was recommended through our consensus process and therefore, should be considered on a case-by-case basis. Prescribing must be individualized to review potential adverse side effects and medication interactions and discussed with patient before prescribing.

## Medication polypharmacy considerations in cognitive symptoms

The American Geriatrics Society Beers criteria (updated 2019) were initially created as a list of potentially

inappropriate medications for use in older adults, 65 years and older.<sup>44</sup> This guidance on medication effectiveness and safety management may be extrapolated to help PASC patients of all ages by assessing the potential detrimental impact of prescribed medication(s) on cognitive function. The Beers criteria identified medications in the following classes that have shown to have depressed CNS effects such as sedation, confusion, and possible cognitive decline:

1. Anticholinergics
  - a. First-generation antihistamines (e.g., diphenhydramine, hydroxyzine, meclizine)
  - b. Antispasmodics (e.g., scopolamine)
  - c. Central alpha agonists (e.g., clonidine)
2. Central nervous system
  - a. Antidepressants (e.g., amitriptyline, nortriptyline, paroxetine)
3. Antipsychotics - conventional and atypical (e.g., risperidone, quetiapine, olanzapine, ziprasidone)
4. Benzodiazepines (e.g., alprazolam, diazepam, lorazepam, temazepam)
5. Nonbenzodiazepine, benzodiazepine receptor agonist hypnotics (e.g., eszopiclone, zaleplon, zolpidem)
6. Skeletal muscle relaxants (e.g., varisoprodol, cyclobenzaprine)

Of note, these criteria should not be used in isolation to supersede clinical judgment and individualized patient care. When considering the benefits of using pharmacological treatments, the decision to maintain medications must be specific for the individual patient. Deprescribing medications may be beneficial for individuals with PASC-related cognitive symptoms. Decreasing anticholinergic burden in patients with mild cognitive symptoms<sup>45</sup> and has been recommended for the aging brain.<sup>46</sup> It is

important to note that use of pharmacologic agents and supplements varies across PASC clinics. Prescribing medications or supplements should be considered on a case-by-case basis, recognizing the limited scientific evidence. Additionally, there should be consideration of the out-of-pocket cost of supplements, the risk of medication interactions, lack of federal regulation and oversight of supplements, and possible side effects.

## Behavioral health and resilience

It is important to highlight those social stressors associated with the COVID-19 pandemic, including social distancing, lifestyle modifications, and coping with familiar peers or family who are ill, that may adversely affect mental health, which can subsequently affect cognition. All individuals with PASC should be screened for mental health disorders such as anxiety, depression and acute stress disorder or posttraumatic stress disorder to guide appropriate interventions. Life stressors and mental health conditions can also negatively affect cognitive function and thus need to be assessed. Those with PASC who have been self-advocating and engaging in adaptive behaviors with a goal toward improved quality of life should be acknowledged for their psychological resilience in spite of their health concerns. Emergent research has also shown that individuals with stronger social and family support, better sleep quality, and more physical activity demonstrate better resilience in face of the pandemic,<sup>47</sup> underscoring several of our treatment recommendations. Patients with PASC should be encouraged to maintain social engagement and lean on support systems during this challenging period, which may include patient advocacy groups.

## HEALTH EQUITY

In the context of PASC, it is important to focus on health equity, health disparities, and social determinants of health (SDOH). Clinicians are encouraged to reference the Appendix Table A1: Health Equity Considerations and Examples in Post-Acute Sequelae of SARS-CoV-2 Infection (PASC): Cognitive Symptoms for guidance about the integration of health equity considerations with the evaluation and treatment of individuals with PASC. In addition, the Multi-Disciplinary Collaborative Consensus Guidance Statement on the Assessment and Treatment of Fatigue in Post-Acute Sequelae of SARS-CoV-2 Infection (PASC) Patients<sup>25</sup> provides an extensive discussion of health equity considerations. Importantly, data on PASC and health disparities are emerging,<sup>48</sup> and some inequities are related to SDOH.<sup>49</sup> Community organizations have served and will continue to serve essential roles during the pandemic.<sup>50</sup>

The implementation of policies that support the health of all individuals in the United States is also vital.<sup>51</sup> Vulnerable populations facing PASC may have little or no health insurance coverage and may have less job security, limited flexibility in their roles, and diminished entitlement to sick pay and occupational health services.<sup>52</sup>

Telemedicine, including phone calls and virtual visits, may be helpful for ongoing follow-up. Virtual visits also offer additional opportunities for caregiver participation, which can be particularly useful for a patient with cognitive symptoms, because the patient and caregiver do not need to be in the same geographic location.

## FUTURE DIRECTIONS IN ASSESSING AND TREATING COGNITIVE SYMPTOMS RELATED TO PASC

The goal of this PASC Collaborative Guidance Statement is to create a coordinated and systematic approach to the evaluation and treatment of patients presenting with PASC-related cognitive symptoms. The recommendations represent a consensus of a large national multi-disciplinary collaborative of centers focused on the treatment of individuals with PASC. The recommendations are based on the most current available data, extrapolation from evidence in similar conditions, and the combined clinical experience of treating thousands of patients with PASC-related cognitive symptoms.

It is important to recognize emerging knowledge and recognition of cognitive symptoms as a significant outcome of COVID-19. The PASC Collaborative recognizes patients with PASC typically present with a cluster of symptoms that cross multiple body systems and our focus in this statement only touches the surface of the emerging evidence. The duration and recovery trajectories of the PASC-related impairments have yet to be characterized, and some post-COVID patients will experience long-term disabilities. Beyond cognitive symptoms, case studies provide evidence that patients with COVID-19 can develop a range of neurological complications including those arising from stroke, encephalopathies, inflammatory syndrome, microbleeds and autoimmune responses.<sup>53</sup> Future PASC Collaborative Consensus Guidance Statements will further examine and offer guidance on mental health and neurology symptoms in patients with PASC.

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## DISCLOSURE

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## APPENDIX

**TABLE A1** Health equity considerations and examples in postacute sequelae of SARS-CoV-2 infection (PASC): Cognitive symptoms

| Category                                                                                                                                                                                                                        | Comment                                                                                                                                        | What is known                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Clinical considerations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Racial/ethnic minority groups<br><i>Example: People who identify as Black (including African-American), American-Indian/Alaska Native, Pacific Islander, Asian-American, and Mixed Race, and/or Latino/Hispanic (ethnicity)</i> | Racial/ethnic minority groups may have multiple factors that affect their overall cognitive scores.                                            | Cognitive testing may be biased or inaccurate in people who identify with racial/ethnic minority groups or other marginalized groups owing to a variety of issues such as not having English as a primary language, not properly accounting for variations in dialect or education level, and stereotype threat. <sup>HE-CI-1</sup> Stereotype threat refers to an important psychological phenomenon in which concern of following negative stereotypes about one's group impairs task performance. <sup>HE-CI-2</sup> | Stereotype threat is common, particularly among non-White students, and interferes with learning. <sup>HE-CI-2</sup> Clinicians administering cognitive testing that may elicit stereotype threat should be aware of factors that may affect test scores and clinical recommendations.                                                                                                                                                                                                                                                                                                                                                     |
| Biological sex<br><i>Example: Female Adults</i>                                                                                                                                                                                 | Physiologic and biologic sex differences should be considered for both the diagnosis and treatment of PASC-related cognitive dysfunction.      | Pregnant women frequently have pregnancy-related fatigue, and they may be at higher risk for more severe COVID-19 infections and symptoms, particularly women who have certain comorbidities and other characteristics (eg, older age, diabetes, kidney disease, obesity). <sup>HE-CI-13</sup>                                                                                                                                                                                                                          | Pregnant or postpartum women may experience fatigue, insomnia, and cognitive dysfunction that can be challenging to distinguish from PASC-related cognitive symptoms. Pregnant women may need alternatives to diagnostic testing (eg, computed tomography due to radiation exposure) to avoid potential harm to the fetus. The risks and benefits of medications and other treatment interventions should be assessed for both mother and fetus. Exercise prescriptions may be impacted by symptoms such as excessive vomiting and weight loss in the first trimester and large girth, back pain, or pre-eclampsia in the third trimester. |
| Gender<br><i>Example: Women</i>                                                                                                                                                                                                 | There may be differences in the reporting of PASC-related symptoms by gender.                                                                  | Women may report PASC-related symptoms more than men, with a common symptom being cognitive impairment. <sup>HE-CI-3, HE-CI-4, HE-CI-5</sup>                                                                                                                                                                                                                                                                                                                                                                            | More studies are needed to better determine the incidence and prevalence of post-COVID cognitive symptoms across the gender spectrum (eg, men, women, nonbinary and gender nonconforming identities). Additional PASC-related findings of fatigue, insomnia, and mood disorders may also contribute to cognitive symptoms.                                                                                                                                                                                                                                                                                                                 |
| Age<br><i>Example: Older individuals</i>                                                                                                                                                                                        | Healthy older individuals as well as those with mild cognitive symptoms and dementia have among the highest rates of COVID-19 infection. Older | Cognitive symptoms in healthy older individuals as well as those with mild cognitive impairment or dementia may be due to direct or indirect                                                                                                                                                                                                                                                                                                                                                                            | Cognitive testing should be conducted by clinicians experienced in working with older individuals. A review of all medications (including                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

(Continues)

TABLE A1 (Continued)

| Category                                                                                                                                                                                                                                           | Comment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | What is known                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Clinical considerations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                    | people may actually have or be perceived as having cognitive symptoms.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | effects of the virus. <sup>HE-CI-6</sup> In addition to PASC-related cognitive symptoms, older individuals are at higher risk for preexisting cognitive conditions such as Alzheimer's disease, stroke, and Parkinson's disease. Delirium is also more common in older people. Polypharmacy and inappropriate dosing may also affect cognition. Age-related disability such as impaired vision or hearing may affect cognitive testing.                                                                                                                                                         | over the counter drugs and supplements) is critical in this population. Vision and hearing should be checked and for correctable impairments and interventions should be prescribed. Screening for depression and sleep disorders is appropriate. Alcohol and other substance use should not be overlooked in this population as it may contribute to cognitive symptoms.                                                                                                                                                                                                                                                                                                                                            |
| Environmental exposure<br><i>Example: People who live in low socioeconomic environments that expose them to various types of environment-related stressors</i>                                                                                     | Social determinants of health (SDOH) include the conditions in which people are born, work, live, and age and the wider set of forces and systems that shape the conditions of daily life. SDOH often focus on environmental exposures that may negatively impact cognitive function which may be exacerbated by PASC-related cognitive symptoms.                                                                                                                                                                                                 | Both children and older individuals exposed to air pollution exhibit signs of cognitive dysfunction. <sup>HE-CI-7</sup> Several modifiable risk factors have been found to have a higher risk of cognitive function in the form of Alzheimer's disease and related dementias (ADRDs): neighborhoods that are economically underinvested with insufficient physical resources, lower food security in early and late life, lower socioeconomic status, less education, manual labor for employment, racial discrimination, higher levels of stress, and early-life adversity. <sup>HE-CI-8</sup> | The incorporation of healthy lifestyle practices is a strong recommendation for physical and cognitive health. Recovery from PASC-related cognitive dysfunction may be impacted by exposure to the natural and psychosocial environment in which people live. Clinicians should be aware of environmental-related stressors, and provide interventions that aim to decrease them whenever possible. Referrals to a local food bank and other community resources can be very helpful to individuals. People knowledgeable about these types of resources include, but are not limited to, medical social workers and community leaders. Appropriate referrals can facilitate physical and cognitive health recovery. |
| Disability<br><i>Example: People who have impairments in physical and/or mobility, psychological or mental health, vision, hearing, emotional or social relationships, cognitive or learning, speech and communication, and other disabilities</i> | When possible, cognitive rehabilitation testing and treatment strategies should be adapted to accommodate one's primary disability. Among persons with disability, many have underlying cognitive impairments. Some of these are static (eg, cerebral palsy, traumatic brain injury, stroke) and some are progressive (eg, dementia, multiple sclerosis). These pre-morbid health conditions may contribute to more profound PASC-related cognitive dysfunction and complicate the diagnosis and treatment of PASC-related cognitive impairments. | During the SARS-CoV-2 pandemic, people with intellectual disability and/or cognitive impairments may require assistance from their caregivers to understand changing recommendations regarding health advisories, social practices and their personal safety. <sup>HE-CI-9</sup> People with intellectual disability and PASC-related cognitive dysfunction are susceptible to fraudulent exploitation by others. <sup>HE-CI-9</sup>                                                                                                                                                            | The compounding of a chronic disability with PASC-related cognitive symptoms may have an additional impact on function and community participation. Before PASC, individuals with chronic visual deficits, hearing deficits, or communication deficits may rely on intact cognition for memory, attention, multitasking, problem solving, and communication strategies. Individuals with PASC-related cognitive symptoms may report more difficulty with their primary disability. Recommendations should be customized to the specific impairments and individual.                                                                                                                                                  |

(Continues)

TABLE A1 (Continued)

| Category                                                                                                                     | Comment                                                                                                                                                                                                                                           | What is known                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Clinical considerations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Religion<br><i>Example: People who identify with a shared belief in what is sacred, holy, divine, spiritual, or reverent</i> | Participation in religious/spiritual based practices may contribute to better cognitive and mental health during the COVID-19 pandemic; however, due to social restrictions many people were not able to attend their usual religious gatherings. | Positive associations have been reported between religion and spiritual based practices and may be protective in middle aged and elderly adults. <sup>HE-CI-10</sup> As a result of the COVID-19 pandemic, spiritual/religious leaders and support groups are attempting to assist with an increased need to attend to widespread suffering in the form of loneliness and isolation. <sup>HE-CI-11</sup> At the same time, these leaders have a responsibility to keep people as safe as possible and avoid spread of the virus. | If in alignment with personal beliefs and in an appropriately safe manner, PASC survivors should be encouraged to return to and/or pursue involvement in religious/spiritual practices as another form of social connection and community support. Caution should be recommended for cognitively impaired individuals as they may be more susceptible to not adhering to safety precautions as well as predatory practices. Clinicians should also recognize that many individuals may receive similar health benefits from nonreligious/nonspiritual types of social interaction such as joining a nature group. |
| Immigration<br><i>Example: People who have come from another country in order to live in the United States</i>               | Clinicians should be aware that people who immigrated may encounter health equity issues related to communication and/or education that may impact both cognitive evaluation and treatment.                                                       | Laws at the local, state, and federal level influence immigrants' access to health insurance and medical care, cash assistance, food assistance, education, and other vital services. <sup>HE-CI-12</sup>                                                                                                                                                                                                                                                                                                                        | Treatment for PASC-related cognitive symptoms may require customized language and cultural considerations to achieve health equity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

*Note:* This table is included in the Appendix to provide additional information for clinicians who are treating patients for PASC-related cognitive symptoms. This is not intended to be a comprehensive list, but rather to provide clinical examples as they relate to health equity, health disparities, and social determinants of health. The literature demonstrates that all marginalized groups face socioeconomic barriers and access to care barriers, though these may or may not be barriers for a specific individual patient. People with intersectional identities (eg, those who identify with more than one underrepresented or marginalized group), often face enhanced levels of bias and discrimination.

Abbreviation: PASC, postacute sequelae of SARS-CoV-2 infection.

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